

Recovering Sparse Neural Connectivity from Partial Measurements: A Covariance-Based Approach with Granger-Causality Refinement

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Problem

Goal: Recover full $N \times N$ connectivity matrix W from **partial measurements** across multiple recording sessions.

Challenge: In small brain circuits (*C. elegans*, zebrafish, organoids), we can only observe a *subset* of neurons per session.

Dynamical model:

$$x_{t+1} = W\phi(x_t) + b_t$$

Each session k observes $\mathcal{M}_k \subset \{1, \dots, N\}$ and stimulates S_k .

Key insight: Pairwise covariance statistics can be **accumulated across sessions** whenever neurons i, j are co-observed.

Method

Step 1 — Covariance accumulation over $K=50$ sessions:

$$\hat{\Sigma}_{x_{t+1}, x_t}(i, j) = \frac{1}{n_{ij}} \sum_{k: i, j \in \mathcal{M}_k} \hat{\Sigma}^{(k)}(i, j)$$

Step 2 — Linear estimator:

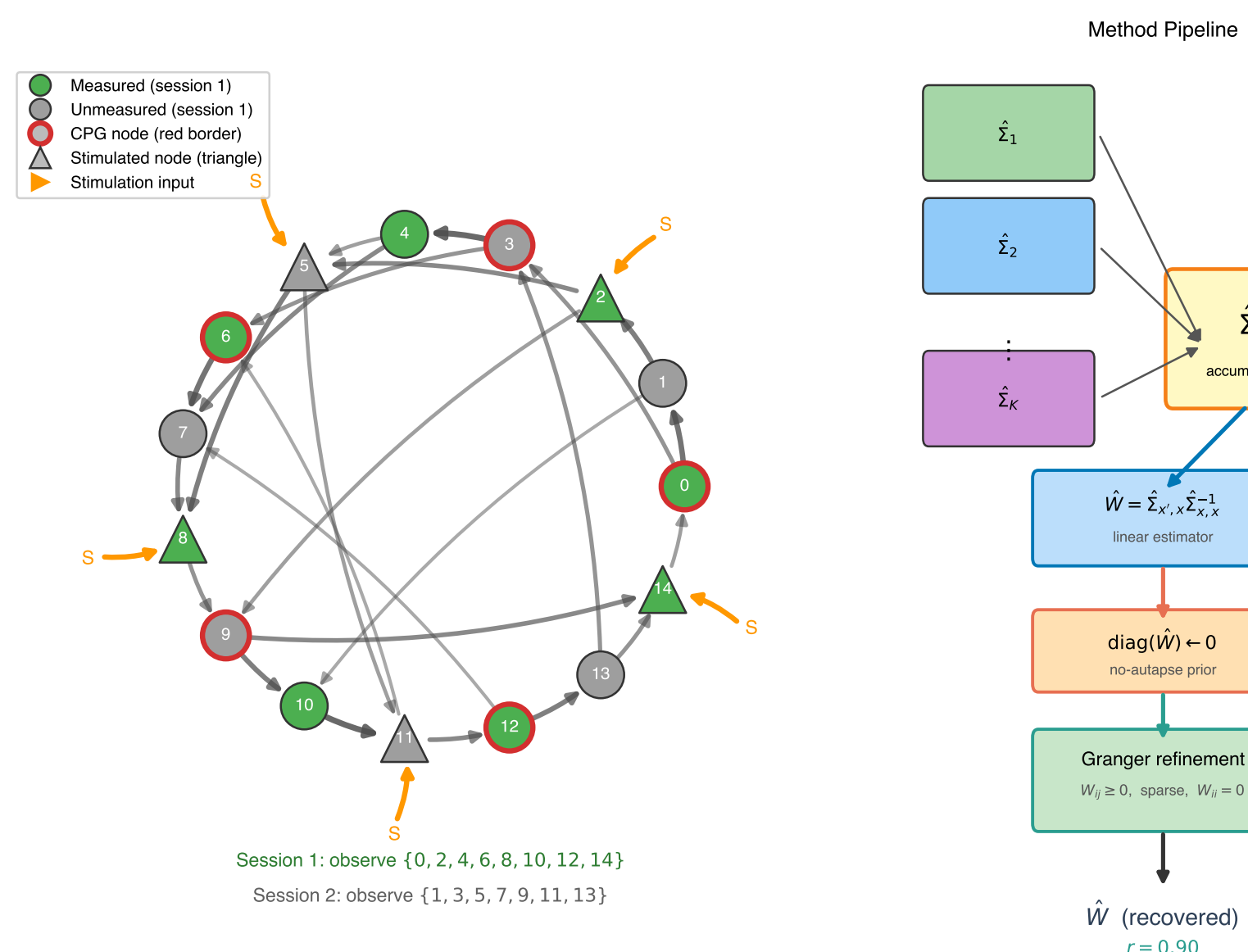
$$\hat{W} = \hat{\Sigma}_{x_{t+1}, x_t} \hat{\Sigma}_{x_t, x_t}^{-1}$$

with structural priors: diagonal zeroed ($W_{ii}=0$, no autapses) and non-negativity ($W_{ij} \geq 0$).

Step 3 — Granger-causality refinement: Projected gradient descent enforcing Granger non-causality zeros ($W_{ij} = 0$ where contemporaneous covariance exceeds lagged).

Identifiability: $K \geq \log(N^2/\delta)/p^2$ sessions suffice for all-pairs coverage with probability $\geq 1-\delta$.

Pipeline



Key Results

Weight correlation $r = 0.90$ (Granger-refined)

Improvement over prior 31% (0.083 vs. 0.120)

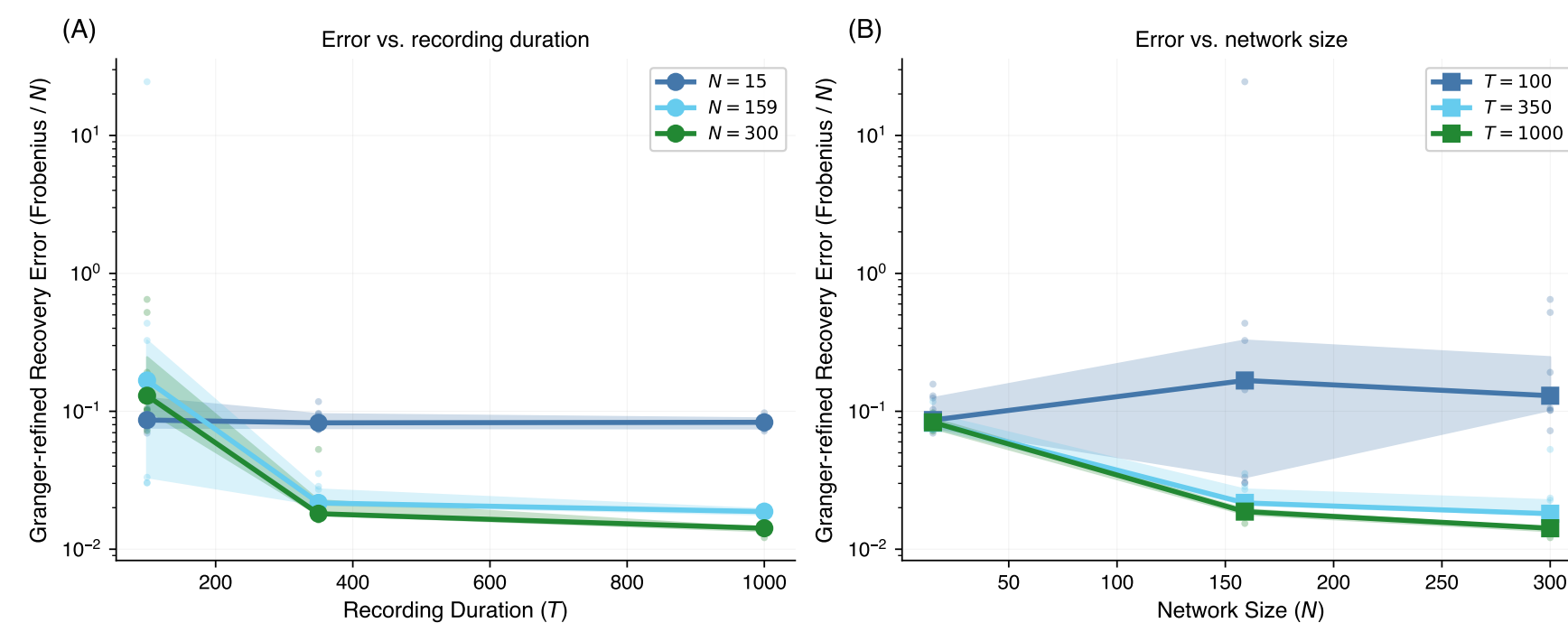
Best recovery $N=300, T=1000$: error 0.014

Sampling Sufficiency

With $m = \lceil 2N/3 \rceil$ measured/session, $P_{\text{pair}} = m(m-1)/(N(N-1)) \approx 0.44$. For $K=50$: $\Pr[\text{any pair missed}] \leq \binom{N}{2} (0.56)^{50} \approx 0$. **Coverage is not the bottleneck.**

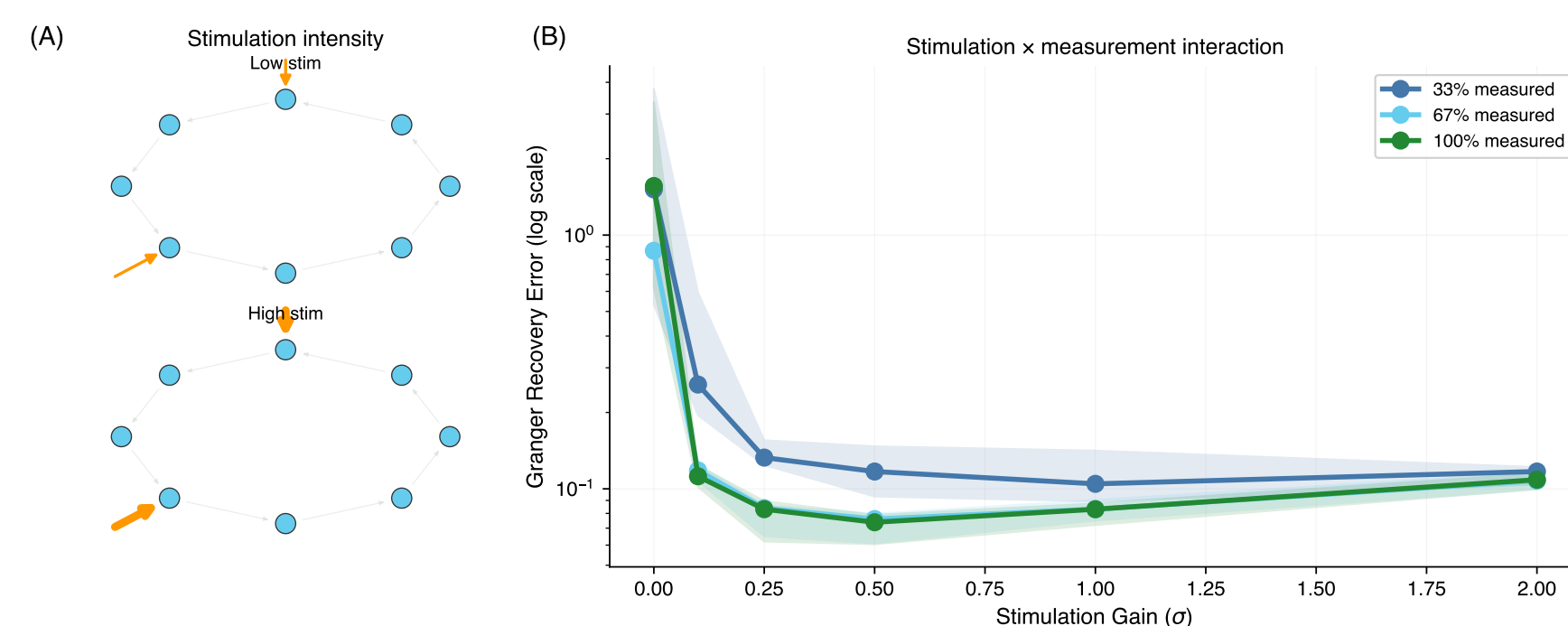
Estimation quality depends on the **DOF ratio** KT/N^2 : at $N=300, T=100$, $\text{DOF}=0.56$ (underdetermined $\Rightarrow$ high variance).

Scaling (E1)



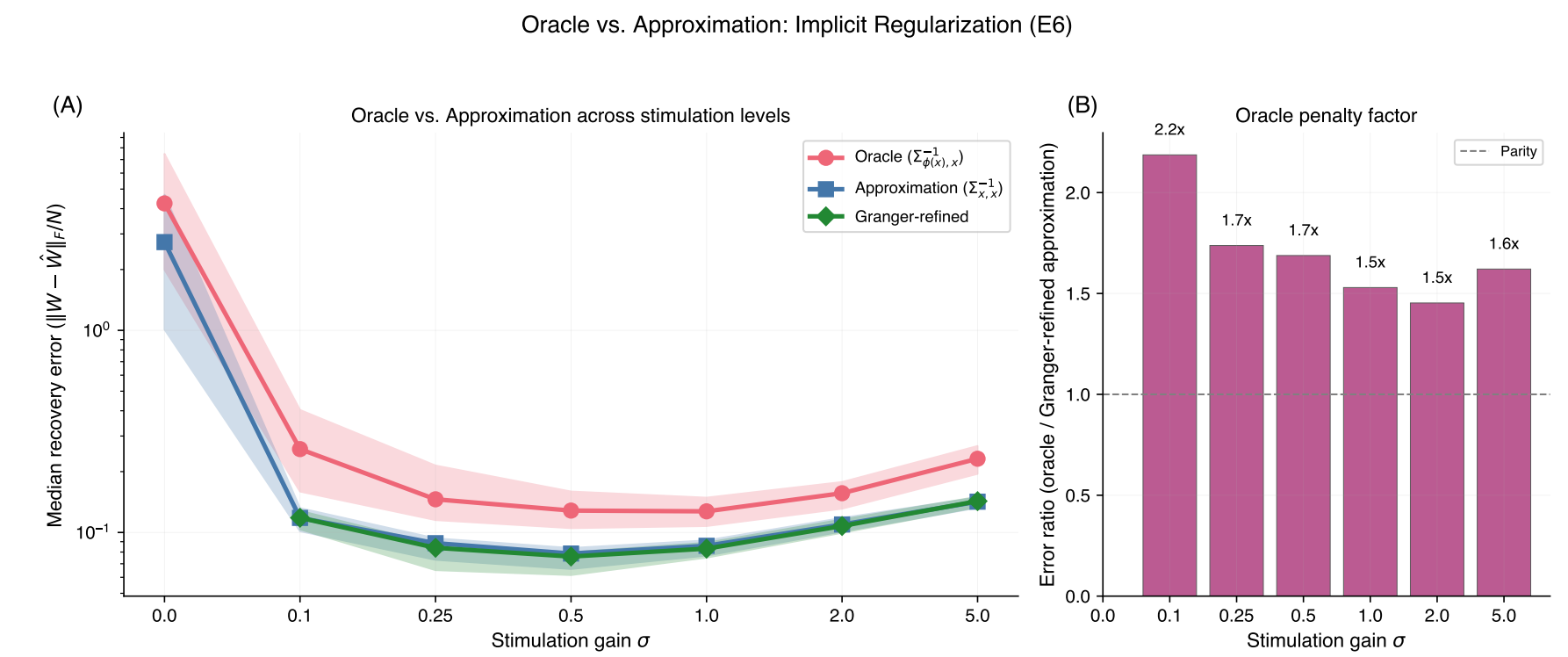
Error decreases with T for all network sizes. Best: $N=300, T=1000$ achieves error 0.014 (97.5% over chance). Dots: individual topologies ($n=10$).

Control–Estimation Tradeoff (E3)



Stimulation is **necessary** for identifiability but **excessive** noise degrades recovery. Optimal: $\sigma \approx 0.5-1.0$ at 66%+ measurement.

Implicit Regularization (E6)



The “incorrect” linear approximation **outperforms the natural plug-in oracle** that knows the nonlinearity at *all* tested regimes: a James–Stein phenomenon driven by oracle ill-conditioning.

Why? $\Sigma_{\phi(x), x} = D\Sigma_{x, x}$ where tanh compression makes D ill-conditioned, amplifying noise in the oracle inversion.

Practical implication: No need to characterize neuronal nonlinearity: the simpler model is *better*.

Error Decomposition

Dominant bottleneck: CPG–state correlation, **not** model mismatch.

Error source	$\ E\ _F/N$	Fraction
Model mismatch (e_1)	0.080	1×
CPG correlation (e_2)	0.225	2.8×
Total estimation	0.087	—

$\Rightarrow$ Future gains from modeling **intrinsic dynamics**, not refining the nonlinear approximation.

Practical Guidelines for Experimentalists

- 1. Measurement:** $\geq 50\%$ neuron coverage per session
- 2. Stimulation:** Moderate ($\sigma \approx 0.5-1.0$) on $\sim 33\%$ of neurons
- 3. Sessions:** $K \geq 50$ with random masks for all-pairs coverage
- 4. Model:** Use the linear approximation — simpler *and* better
- 5. Next:** Model CPG dynamics to tackle the $\sim 3\times$ dominant error